

1-е сечение 4-мусса

Бер: 25%

$$U_{\text{нак}} = 27,5 \text{ кВ}$$

Ластовы гравитаци

$$PC-70 \quad D_1 = 255 \text{ мм} \quad H = 127 \text{ мм}$$

$$L_1 = 300 \text{ мм} \quad \lambda_2 = \text{см/кВ}$$

$$\lambda = 0,35 \cdot \left(1 - \frac{L_1 + 4n}{H}\right) = 0,35 \cdot \left(1 - \frac{300}{127}\right)$$

$$k = 1 + 0,5 \cdot \left(\frac{L_1 + 4n}{D} - 1\right) = 1,08$$

$$n \geq \lambda_2 \cdot U_{1, \text{max}} / L_{3\varphi} = \frac{3,5 \cdot 40,5}{300} = 4,725 \approx 5$$

$$\lambda_2 = \frac{L_{3\varphi}}{U_{1, \text{max}}} = \frac{277,7}{40,5 \cdot 10} = 68,56$$

$$\Rightarrow L_{3\varphi} = U_{1, \text{max}} \cdot \lambda_2 = 141,75 \text{ мм} = 1417,5 \text{ мм}$$

2-е сечение 4-мусса

$$h = 12 \text{ мм}$$

$$S = 70 \text{ мм}^2$$

$$L = 25 \text{ мм}$$

$$U_H = 10 \text{ кВ}$$

$$C_0 = \frac{2\pi \epsilon_0}{\ln \frac{2h}{r}} = 650 \cdot 10^{-12} \text{ ф/м}$$

$$I_3 = 3 \omega \cdot C_0 \cdot L U_H = 1,35 \text{ А}$$

$$\omega = 2\pi \nu = 314$$

$$U_H = \frac{10}{\sqrt{3}} = 5770 \text{ В}$$

$$\omega L = \frac{1}{3 \omega C_{11}} \Rightarrow C_{11} = \frac{1}{3 L \omega^2} = 1,35 \cdot 10^{-10}$$

$$3 \omega C_{11} U_{10} = \frac{U_{10}}{\omega L}$$

$$Q = U_H \cdot I_3 = 7289,5 \text{ ВАр}$$

3-ecan

$$U = 20 \text{ kV}$$

$$I = 15 \text{ kA}$$

$$h = 14 \text{ m}$$

$$f = 5 \text{ m}$$

$$a = 10 \text{ m}$$

$$U_{\text{reg}} = h \cdot \frac{h_{cp}}{a} \cdot I_m$$

$$h_{cp} = h - \frac{2}{3} f = 14 - \frac{2}{3} \cdot 5 = 10,6$$

$$U = k \cdot \frac{h_{cp}}{a} \cdot I \Rightarrow 20 = \frac{10,6}{10} \cdot 15 = 318$$

$$U = \frac{318}{20} = 15,9$$

4-цен

4-участка

Мак-Согг name paragraph you-
уас $S_3 = 6 \text{ м}$ $L_3 = 18 \text{ м}$ $h_3 = 6 \text{ м}$

Отрасль: Печер

$P = 30 \text{ Оу.м}$

Мастер мпм - Мерисемат мпм:

$B = \underline{II}$

$$N = ((S_3 + 6 h_3) \cdot (L_3 + 6 h_3) - 7,7 \cdot h_3^2) \cdot n$$

$$10^{-6} = 0,0$$

$$n = 2 \text{ км}^2$$

Самая - I $B = \underline{II}$

уимат - A

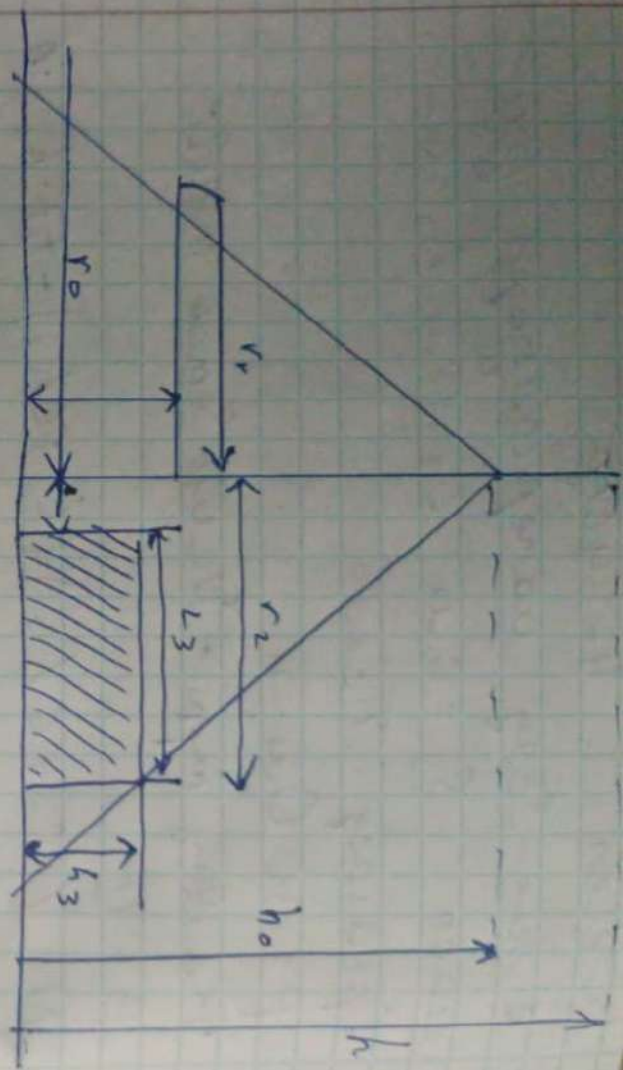
$$100 < P \leq 1000 \text{ м}$$

$$S_6 = 2 + P \cdot 0,01 = 2 + 30 \cdot 0,01 = 2,3 \text{ м}$$

$$CM = 30$$

$$h_0 = 0,85 \cdot h = 25,5$$

$$\Gamma_0 = h \cdot (1,1 - 0,002 \cdot h) = 31,2$$



5-ден 4-мүчкө

$$Z_1 = 160 \text{ Ом}$$

$$Z_2 = 440 \text{ Ом}$$

$$T_1 = 0,85 \text{ мкс}$$

$$U = 475 \text{ кВ}$$

$$U_{\text{min}} = 150 \text{ кВ}$$

$$U_0(t) = U(e^{-t/T_1} - e^{-t/T_2})$$

$$t_{\text{max}} = \frac{T_1 T_2}{T_1 - T_2} \ln \frac{T_1}{T_2}$$

$$\frac{T_1}{T_2} = \frac{52,5}{0,85} = 61,8 \quad \ln = 4,12$$

$$t_{\text{max}} = 0,86 \cdot 4,12 = 3,55 \approx 436 \text{ кВ}$$

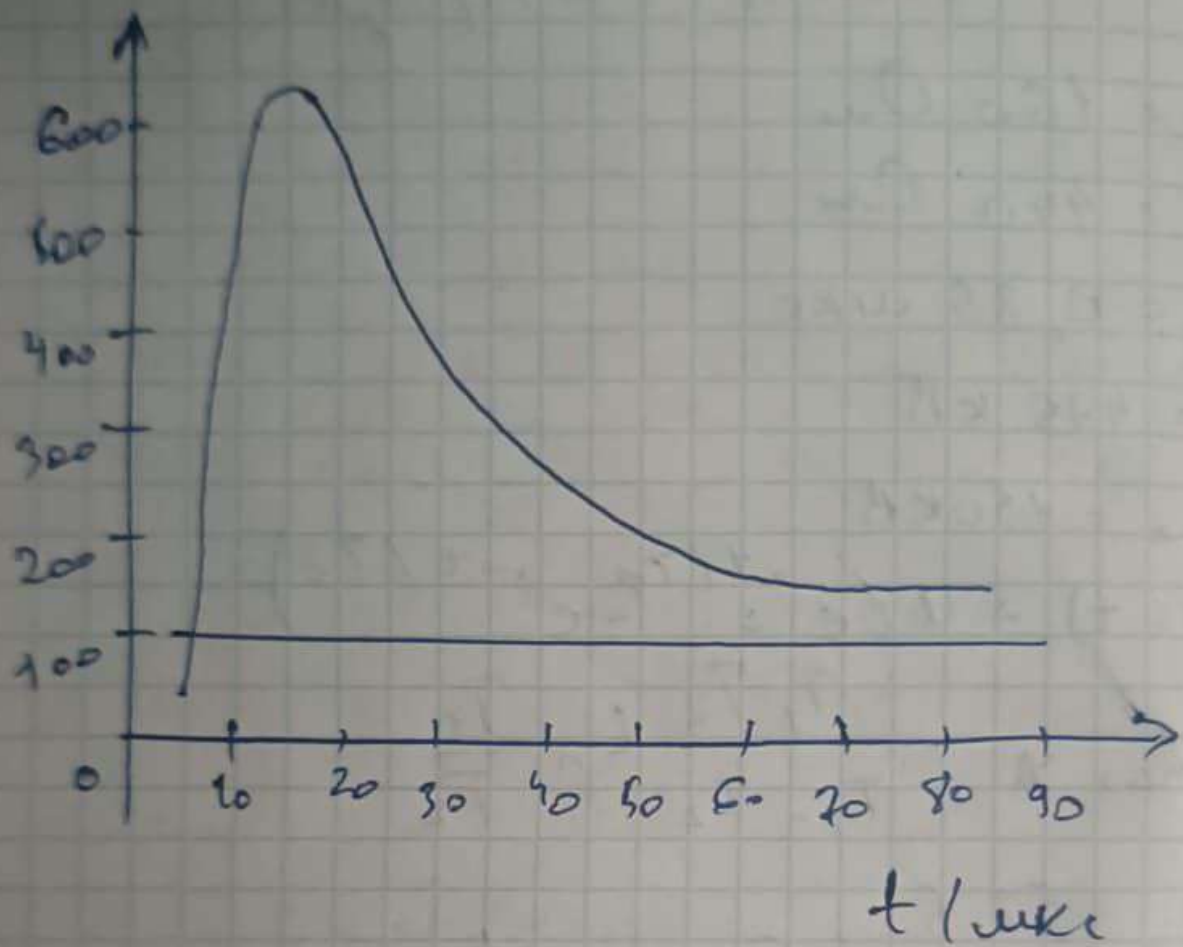
$$t_{0,3} = 0,3 \text{ мкс}$$

$$t_{0,5} = 1,9 \text{ мкс}$$

$$t_{0,8} = 1,6 \text{ мкс}$$

$$t_{\text{max}} = 40 \text{ мкс}$$

$$k = \frac{2 \cdot 2 \cdot 2}{2_1 + 2_2} = \frac{880}{600} = 1,47$$



σ - есеп

U - кыска

$$U = 35 \text{ кВ}$$

$$E_x = \frac{Q}{\epsilon_0 \epsilon_r 2\pi r_x L} =$$

$$R = 24 \text{ мм}$$

$$r = 3 \text{ мм}$$

$$560 \text{ кВ/м}$$

$$\epsilon = 1100$$

$$Q = C \cdot U = 2,366 \cdot 10^{-3}$$

$$\epsilon_r = 2,3$$

$$C = \frac{2\pi \epsilon_0 \epsilon_r L}{\ln \frac{R}{r}} = 6,78 \cdot 10^{-8} \text{ ф}$$

$$\varphi(x) = U \frac{\ln(\frac{R}{x})}{\ln(\frac{R}{r})} = 35 \text{ кВ}$$

$$E = f(x)$$

$$U = f(x)$$

